# Control System Homework 01

Unit 2: Dynamic Models

Problem 3 (2.19): Capacitor Microphone

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This note models the capacitor microphone in Fig. 1, answers (a)–(c), and shows why writing the electric force as $f_e \operatorname{sgn}(\dot{x})$ is not physical.

## 1 What the system is

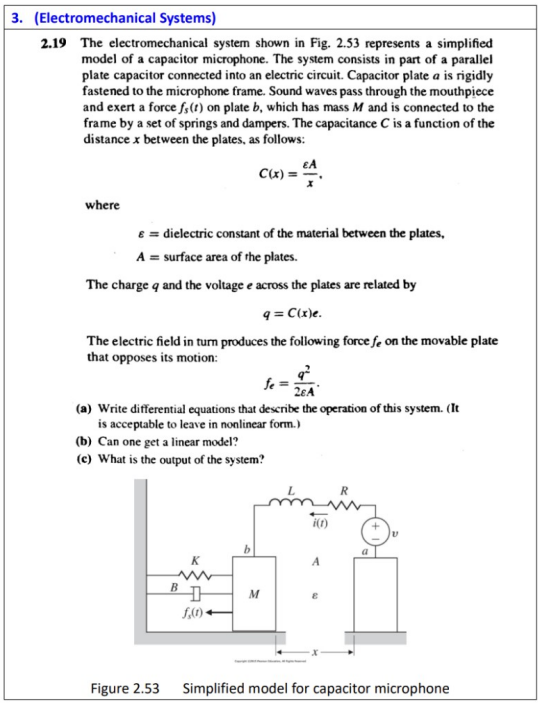
Plate $a$ is fixed. Plate $b$ (mass $M$) is held by a spring $K$ and a damper $B$. Sound applies a force $f_s(t)$ on $b$. The two plates form a capacitor of gap $x$, wired in series with $L$, $R$, and a source $v$.

The problem gives

$$C(x) = \frac{\varepsilon A}{x}, \qquad q = C(x)\, e, \qquad f_e = \frac{q^2}{2\varepsilon A}. \quad (1)$$

A smaller gap means a larger capacitance. $f_e$ is the electric force on the moving plate. The text says this force "opposes its motion."

Coordinate: $x$ is the gap. Increasing $x$ means $b$ moves left and the plates separate. $f_s > 0$ increases $x$. The source is a DC bias $V_0$ (a real condenser microphone also needs bias to turn motion into an electrical signal).


3. (Electromechanical Systems)

**2.19** The electromechanical system shown in Fig. 2.53 represents a simplified model of a capacitor microphone. The system consists in part of a parallel plate capacitor connected into an electric circuit. Capacitor plate $a$ is rigidly fastened to the microphone frame. Sound waves pass through the mouthpiece and exert a force $f_s(t)$ on plate $b$, which has mass $M$ and is connected to the frame by a set of springs and dampers. The capacitance $C$ is a function of the distance $x$ between the plates, as follows:

$$C(x) = \frac{\varepsilon A}{x},$$

where

$\varepsilon$ = dielectric constant of the material between the plates,

$A$ = surface area of the plates.

The charge $q$ and the voltage $e$ across the plates are related by

$$q = C(x)e.$$

The electric field in turn produces the following force $f_e$ on the movable plate that opposes its motion:

$$f_e = \frac{q^2}{2\varepsilon A}.$$

**(a)** Write differential equations that describe the operation of this system. (It is acceptable to leave in nonlinear form.)

**(b)** Can one get a linear model?

**(c)** What is the output of the system?



Figure 2.53 Simplified model for capacitor microphone

**Figure 1:** Fig. 2.53. Simplified capacitor microphone.

## 2 Which way does $f_e$ point?

This circuit is one capacitor: $b$ carries $+q$, $a$ carries $-q$. The plates always have opposite charge. The force depends on $q^2$, so

$$f_e = \frac{q^2}{2\varepsilon A} \geq 0.$$

Opposite charges attract: $f_e$ always pulls $b$ toward $a$, so it always **decreases** $x$. The direction does not depend on $\dot{x}$. If $q = 0$ the force is zero; it never becomes a repulsion.

The official solution (Fig. 2) instead writes

$$M\ddot{x} + B\dot{x} + Kx + f_e \operatorname{sgn}(\dot{x}) = f_s(t), \quad (2)$$

which reads "opposes its motion" as "always against the velocity." That is friction, not electrostatic attraction.

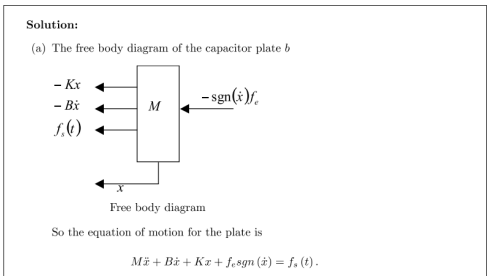



**Figure 2:** Solution FBD: $f_e$ is multiplied by $\operatorname{sgn}(\dot{x})$.

Three problems:

1. **A charged capacitor still pulls at rest.** $\operatorname{sgn}(0) = 0$, so the force vanishes whenever the plate stops.
2. **Closing should attract, not push.** When $\dot{x} < 0$ (gap shrinking), (2) becomes a repulsion.
3. **This force is conservative, not a damper.** Pulling the plates apart takes work; letting them close, the field does work. Power can be positive or negative. sgn makes the power always $\leq 0$.

A better reading of "opposes its motion" is: **you must overcome attraction to separate the plates**, not that the force always points against $\dot{x}$.

So the correct mechanics equation is

$$M\ddot{x} + B\dot{x} + K\big(x - x_{\text{rest}}\big) + f_e = f_s(t), \tag{3}$$

where $f_e$ always shrinks the gap. $x_{\text{rest}}$ is the gap when the spring is unstretched.

## 3 Circuit

KVL around the loop (source, resistor, inductor, capacitor) gives

$$L\dot{i} + Ri + e = V_0, \qquad e = \frac{qx}{\varepsilon A}, \qquad i = \dot{q}. \tag{4}$$

## 4 (a) Nonlinear equations

With state $(x, \dot{x}, q, i)$,

$$\begin{aligned} \ddot{x} &= \frac{1}{M}\Big[f_s(t) - \frac{q^2}{2\varepsilon A} - K\big(x - x_{\text{rest}}\big) - B\dot{x}\Big], \\ \dot{q} &= i, \\ \dot{i} &= \frac{1}{L}\Big(V_0 - Ri - \frac{qx}{\varepsilon A}\Big). \end{aligned} \tag{5}$$

The VPython simulation integrates (5) as is, without linearizing.

## 5 (b) Can we linearize?

Yes, but only near the **biased equilibrium**, for a small sound force.

With $f_s = 0$ and zero velocity and current, the rest point $(x_0, q_0)$ satisfies

$$q_0 = \frac{\varepsilon A V_0}{x_0}, \qquad K(x_0 - x_{\text{rest}}) + \frac{q_0^2}{2\varepsilon A} = 0. \tag{6}$$

The spring must balance a constant attraction, so $x_0 < x_{\text{rest}}$ (the plates sit closer than with no voltage). The sgn model has $f_e = 0$ at rest, so it wrongly puts the equilibrium at $x_{\text{rest}}$.

Let $x = x_0 + \delta x$, $q = q_0 + \delta q$. Dropping second-order terms,

$$\begin{aligned} M\delta\ddot{x} + B\delta\dot{x} + \Big(K - \frac{q_0^2}{\varepsilon A x_0^2}\Big)\delta x + \frac{q_0}{\varepsilon A}\,\delta q = \delta f_s, \\ L\delta\ddot{q} + R\delta\dot{q} + \frac{x_0}{\varepsilon A}\,\delta q + \frac{q_0}{\varepsilon A}\,\delta x = 0. \end{aligned} \tag{7}$$

Electrostatics soften the spring. For a small input, (5) looks like (7); if the input is large the motion is nonlinear and can pull in.

The official solution says the sgn function together with the $q^2$ and $qx$ terms make a useful linear model impossible. These should not be lumped together. $\operatorname{sgn}(\dot{x})$ is not differentiable at $\dot{x} = 0$, so a standard Jacobian linearization fails. The terms $q^2$ and $qx$ are smooth. About a biased point with $q_0 \neq 0$,

$$\delta(q^2) = 2q_0\delta q, \qquad \delta(qx) = q_0\delta x + x_0\delta q,$$

which is exactly (7). Linearizing at $q = 0$ does kill the first-order coupling, and a microphone with no bias would indeed have no useful small-signal output. So dropping sgn is necessary, but one must also expand about $(x_0, q_0)$.

# 6 (c) What is the output?

The output is the current $i(t)$ (or the voltage $Ri$), not the gap $x$. Under bias, $q \approx C(x)V_0$, so

$$i = \dot{q} \approx -\frac{\varepsilon A V_0}{x^2}\,\dot{x}. \tag{8}$$

Thus $i$ tracks **velocity**:

- step: $x$ settles to a new gap, then $i \to 0$ (no DC gain);
- impulse: $x$ rings back; $i$ is nonzero only while the plate moves;
- sine: both $x$ and $i$ keep oscillating.

A capacitor microphone only responds to a *changing* pressure.

# 7 Simulation

We integrate (5) with Euler (semi-implicit on the current). Parameters are scaled so the motion is visible; the equations are the same as in the problem. The input $f_s$ can be a step, a short pulse (impulse), or a sine.

## 7.1 Impulse

Fig. 3: a pulse at $t = 1\,\text{s}$. $x$ leaves equilibrium and returns. $i$ is a pulse, then zero—same shape as $\dot{x}$. The cyan $f_e$ stays almost constant (always an attraction).

## 7.2 Sine

Fig. 4: after $t = 1\,\text{s}$, $f_s$ is sinusoidal. Both $x$ and $i$ oscillate. $f_e$ is almost a flat line; it does not flip when velocity changes sign.

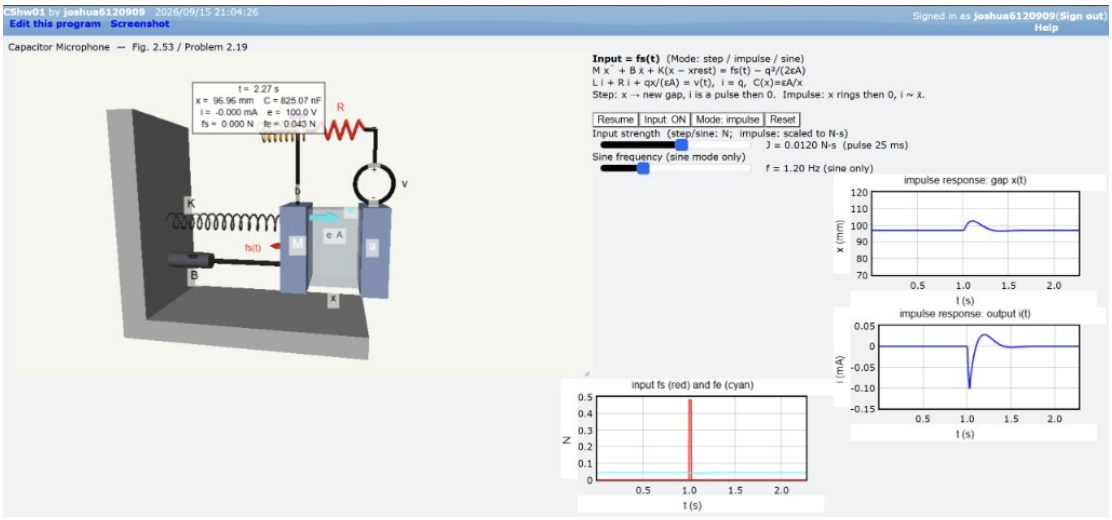



**Figure 3:** Impulse response: $x$ rings back; $i$ is nonzero only during motion.

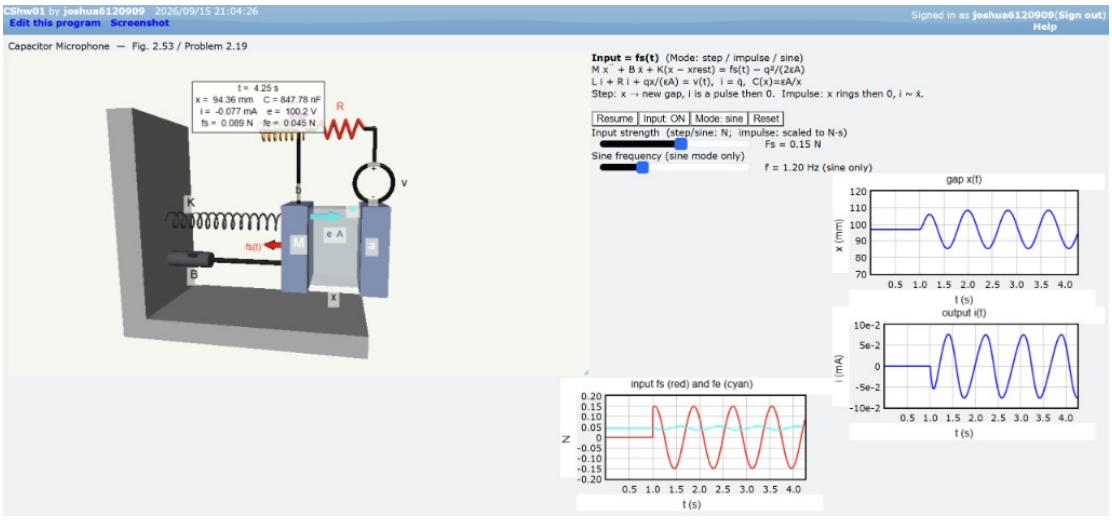



**Figure 4:** Sine input. Red: $f_s$. Cyan: $f_e$ (almost constant). Right: $x$ and $i$.

# 8 Comparison with the sgn model

Same $f_s$, same circuit; only the mechanical force law changes:

$$\text{physics (blue)} \quad M\ddot{x} + B\dot{x} + K(x - x_{\text{rest}}) + f_e = f_s, \tag{9}$$

$$\text{solution (orange)} \quad M\ddot{x} + B\dot{x} + K(x - x_{\text{rest}}) + f_e \,\text{sgn}(\dot{x}) = f_s. \tag{10}$$

Both runs start charged and at rest at $x_{\text{rest}}$. $F_{x,e}$ is the electric force along $+x$ (opening the gap). In physics it should stay negative.

## 8.1 Step: orange has no force at rest

Fig. 5, first 1.5 s with no sound:

- Blue: attraction starts at once; $x$ falls from 100 mm to about 97 mm.
- Orange: $\text{sgn}(0) = 0$, so $F_{x,e} = 0$ and the plate stays at 100 mm.

The plates are charged, yet the orange model does not move. The step is applied only after 1.5 s.

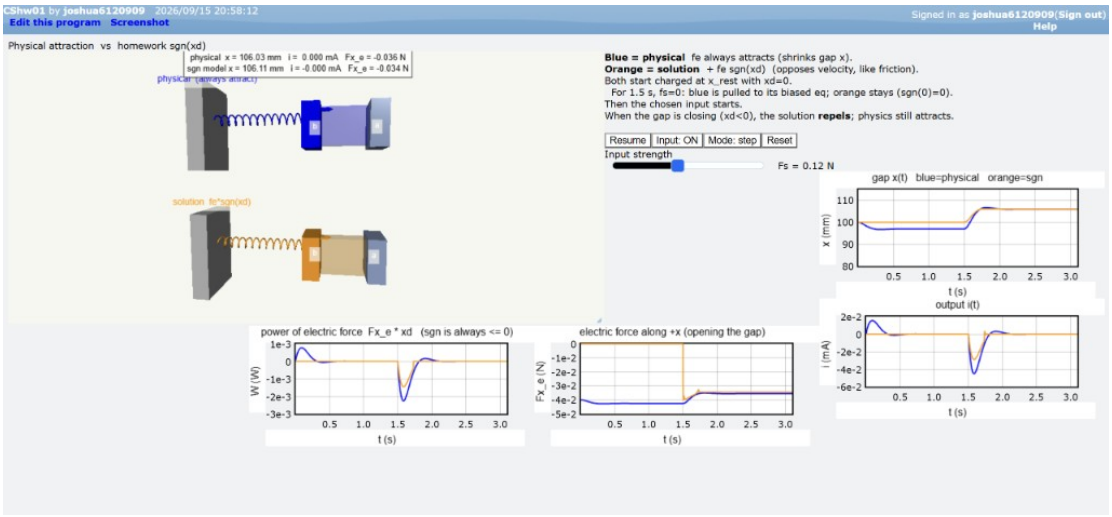



**Figure 5:** Blue = physics, orange = sgn. For $t < 1.5\,\mathrm{s}$ blue is pulled in; orange is not.

### 8.2 Sine: the force should not follow velocity

Fig. 6:

- $x$: blue oscillates about a closer gap; orange stays near 100 mm (no preload).
- $F_{x,e}$: blue is almost a negative DC (always attracting); orange is modulated by $\dot{x}$ and often returns to 0.
- $i$: orange is larger, because the mean gap and the force law are both wrong, so the output is wrong too.

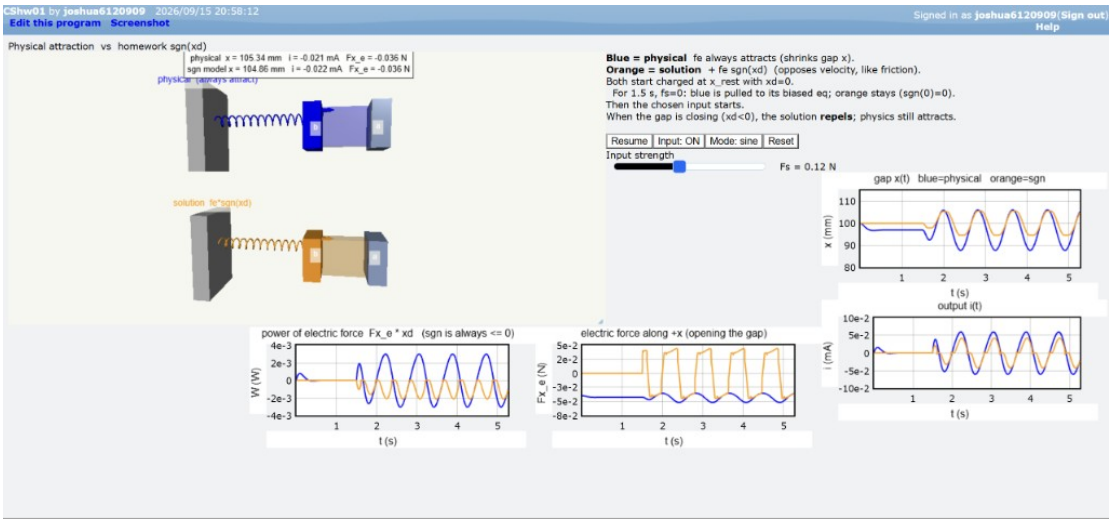



**Figure 6:** Sine comparison. Blue $F_{x,e}$ stays negative; orange tracks velocity, like friction.

A hard sgn also chatters when the speed crosses zero. A true Coulomb attraction should not do that.

## 9 Conclusion

1. **(a)** Use (5): a mass–spring–damper with a one-way attraction, plus an $L$–$R$–$C$ circuit whose $C$ depends on $x$.
2. **(b)** Linearization is possible after dropping sgn, at the biased point $(x_0, q_0)$. $q^2$ and $qx$ are ordinary smooth nonlinearities; the solution overstates that they cannot be linearized.
3. **(c)** The output is $i(t)$. No steady current for a DC pressure; for AC, $i \propto \dot{x}$.

4. $f_e \operatorname{sgn}(\dot{x})$ disagrees with $f_e = q^2/(2\varepsilon A)$: opposite charges always attract. The simulation shows no force at rest in the sgn model, and a velocity-dependent $F_{x,e}$ under sine input.

Read "opposes its motion" as "attraction must be overcome to open the gap." Use (3), not (2).

Programs: `capacitor_microphone.py` (physics), `capacitor_microphone_sgn_compare.py` (comparison).